Lecture 4 recap

In Lecture 4, we:

- Learnt that an nth-order linear ODE is equivalent to a system of n 1st-order linear ODEs
- In general, treat vector-valued, linear ODEs of form

$$\boldsymbol{y}' = \mathbf{A}\,\boldsymbol{y}$$

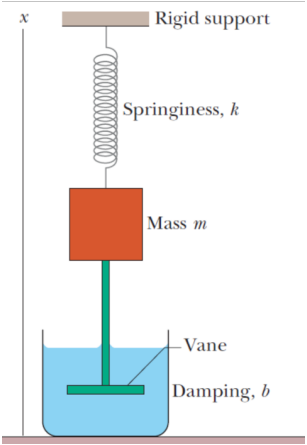
- Discovered a connection between the solution of the homogeneous, constant-coefficient problems with eigenvalues and eigenvectors of a matrix.

Lecture 5

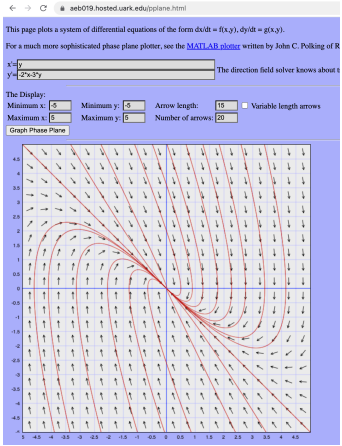
- In Lecture 5, we will:
  1. Learn how to visualise solutions to 2×2 homogeneous systems as trajectories in phase space
  2. Begin understanding how to sketch phase portraits by including critical points and straight line trajectories

How does linear algebra dictate structure of solution space?

Physical space



Abstract phase space



Right-hand figure is from a webpage where you can plot phase portraits

2nd order constant coefficient systems

The mass–spring system is a canonical example of a 2nd-order constant coefficient ODE

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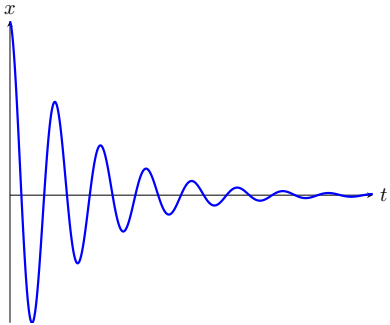
Underdamped case

$$x(t) = \mathrm{e}^{-A\,t}\left(C_1\,\cos(B\,t) + C_2\,\sin(B\,t)\right)$$



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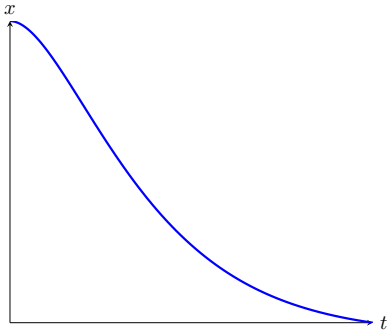
Critically damped case

$$x(t) = (C_1 + C_2\,t)\,\mathrm{e}^{-A\,t}$$



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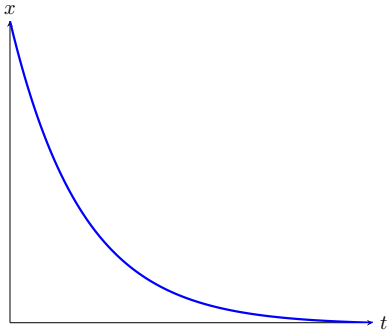

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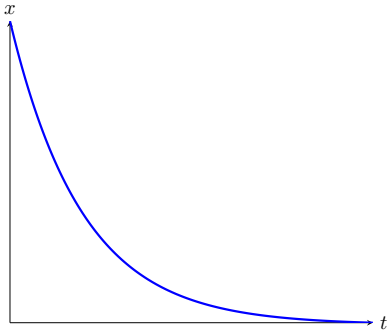

We examine the overdamped case and leave the other cases for self-directed learning.

Exercise 1.12: Which case is the system from Problem 1.2?

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0$$

or $\quad \boldsymbol{y}' = \mathbf{A}\boldsymbol{y} \quad$ where $\quad \boldsymbol{y} = \begin{pmatrix} x \\ v \end{pmatrix} \quad$ and $\quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$

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We construct the phase portrait to establish the structure of solution space.

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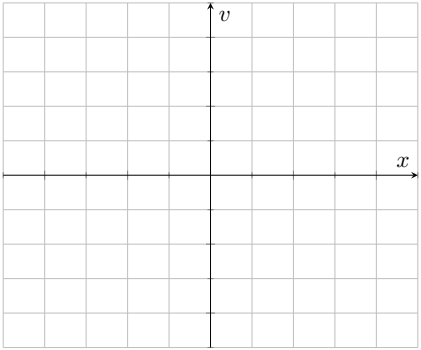
- The xv-plane is called the **phase space**



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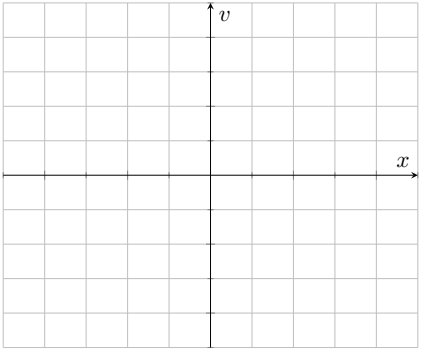

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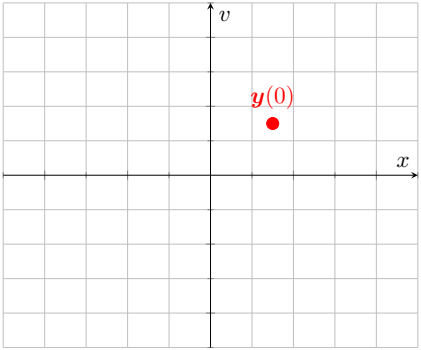

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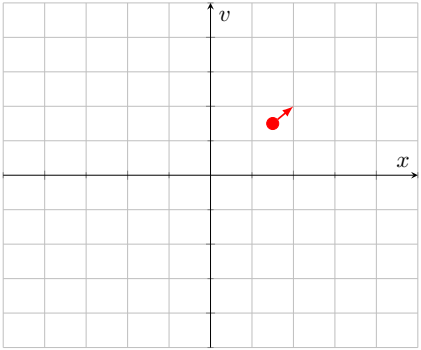

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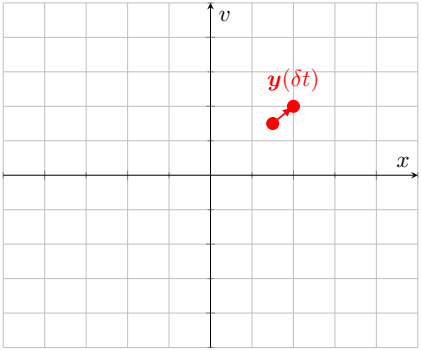

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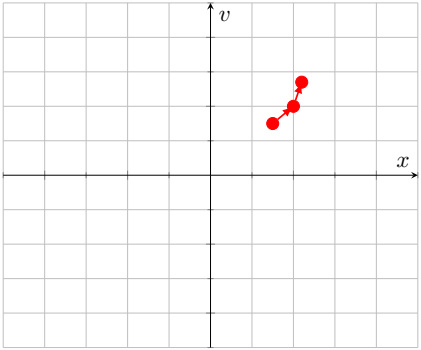

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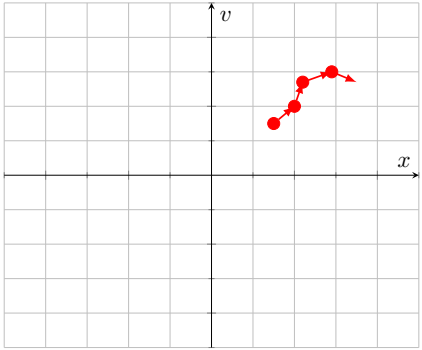

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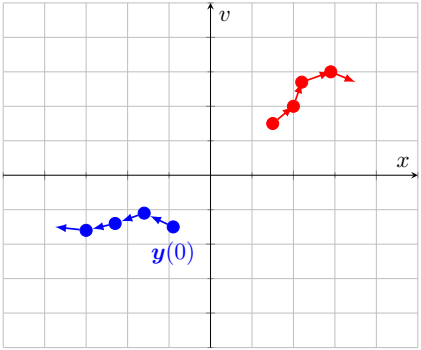

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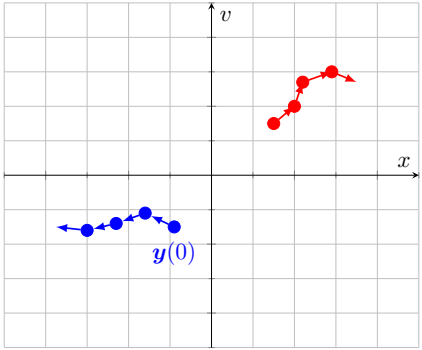
- Uniqueness of IVPs implies that trajectories cannot cross.



Vector field

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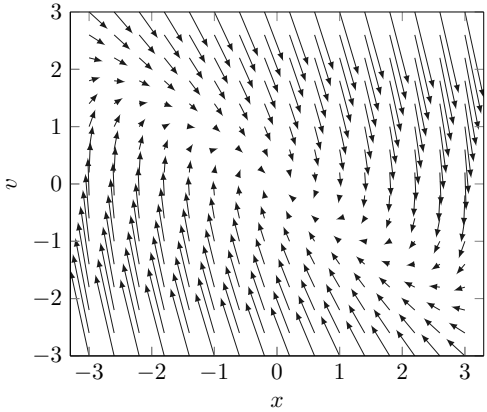

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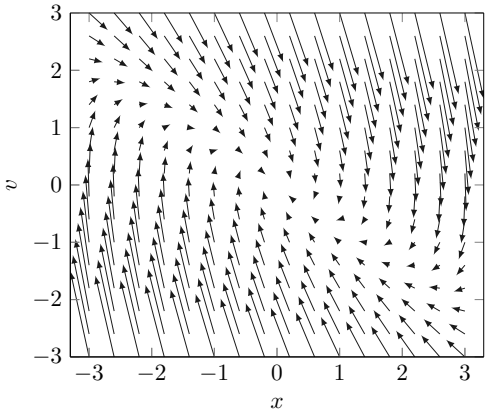
for different points (x, v) in phase space.



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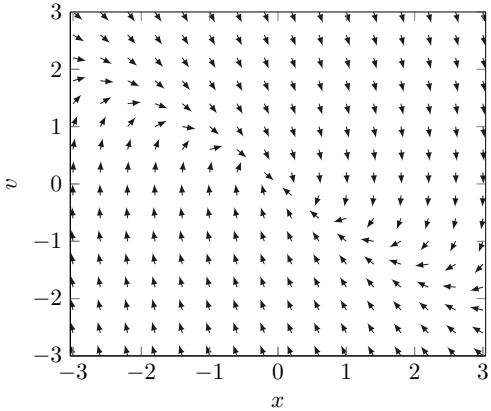

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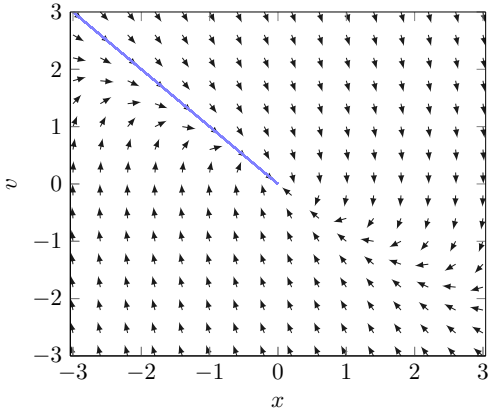

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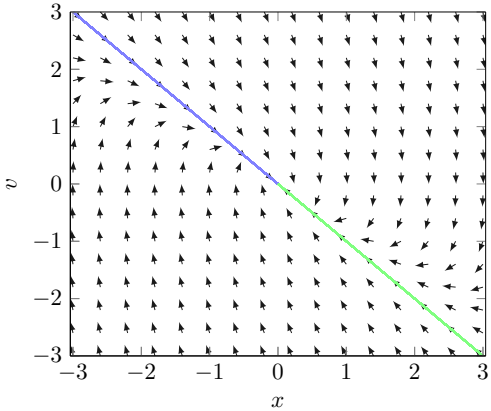

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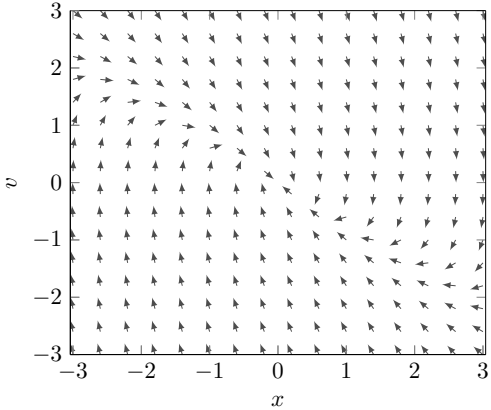
- Only critical point of linear system $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$ for $\mathbf{A}$ invertible is $\boldsymbol{y} = \mathbf{0}$

Phase portraits

- *Phase portrait* is a graphical representation of the solution space



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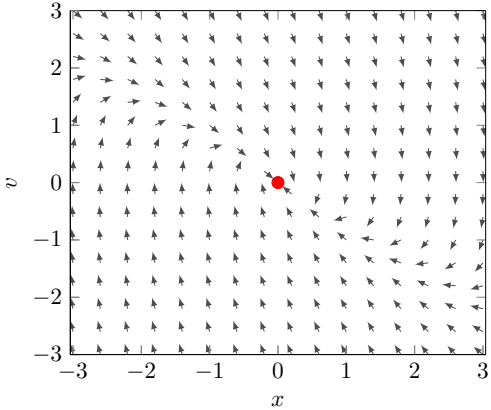

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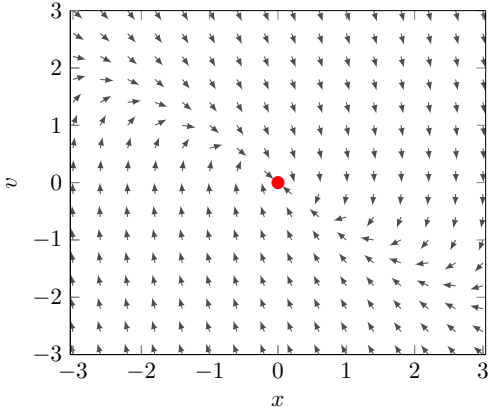

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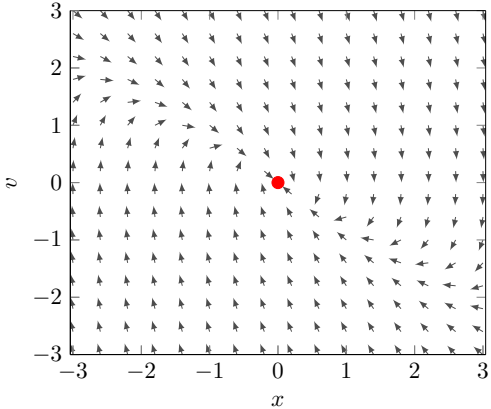
- Contains all critical points
- Contains representative orbits of system
- Matrix **A** alone shapes the structure of the solution space



Sketching phase portraits: Straight line orbits

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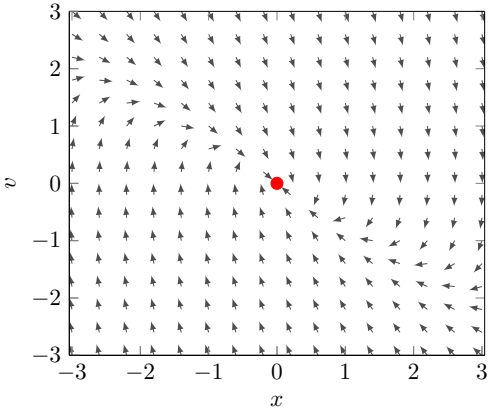

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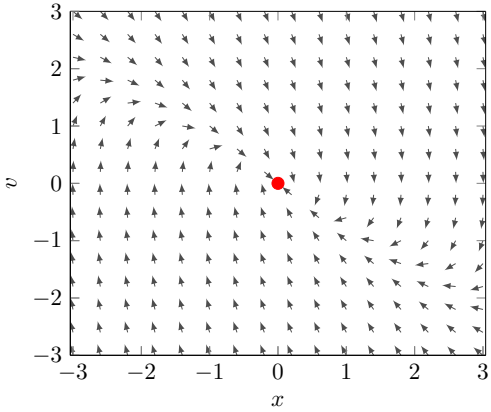
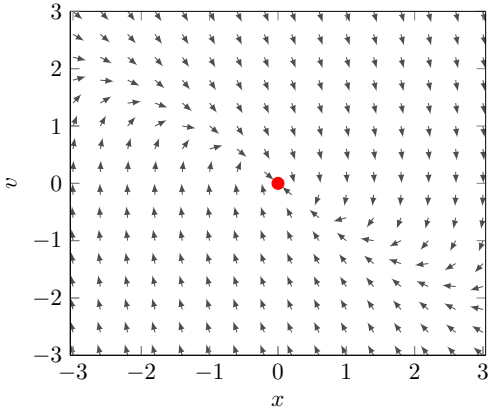


Sketching phase portraits: Straight line orbits

What happens if IC is eigenvector of $\mathbf{A}$?

- Then $\mathbf{A}\boldsymbol{y}_0 = \lambda \boldsymbol{y}_0$
- So $\boldsymbol{y}'$ maintains its direction
  - Except possibly for a flip



Sketching phase portraits: Straight line orbits

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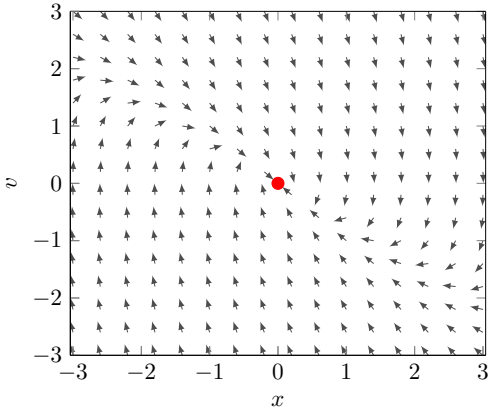

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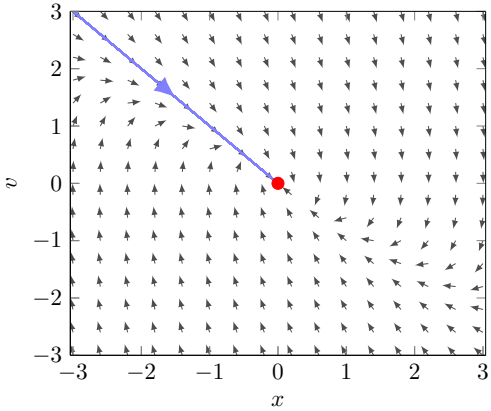

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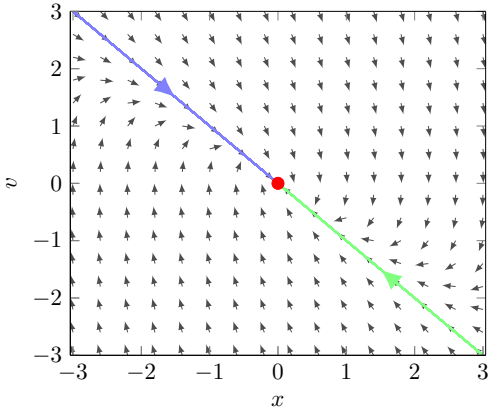

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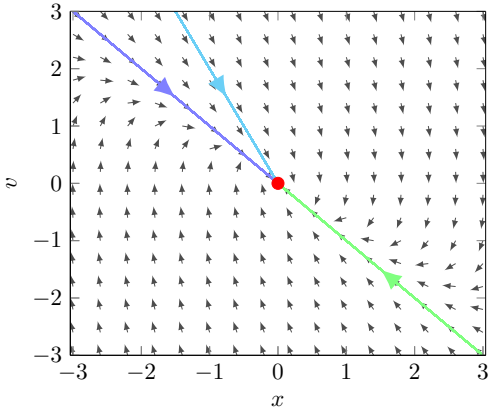

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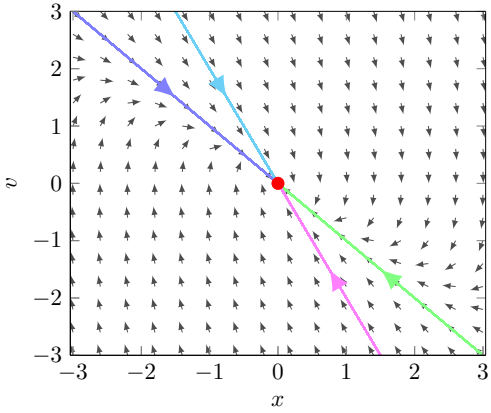
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